

Limit Trigonometri

Pertemuan Ke-6

LIMIT TRIGONOMETRI

Rumus limit fungsi trigonometri

1. $\lim_{x \to 0} \frac{\sin mx}{nx} = \frac{m}{n}$
2. $\lim_{x \to 0} \frac{mx}{\sin nx} = \frac{m}{n}$
3. $\lim_{x \to 0} \frac{\sin mx}{\sin nx} = \frac{m}{n}$
4. $\lim_{x \to 0} \frac{\tan mx}{nx} = \frac{m}{n}$
5. $\lim_{x \to 0} \frac{mx}{\tan nx} = \frac{m}{n}$
6. $\lim_{x \to 0} \frac{\tan mx}{\tan nx} = \frac{m}{n}$
7. $\lim_{x \to 0} \frac{\sin mx}{\tan nx} = \frac{m}{n}$
8. $\lim_{x \to 0} \frac{\tan mx}{\sin nx} = \frac{m}{n}$
9. $\lim_{x \to 0} \frac{\sin x}{x} = 1$
10. $\lim_{x \to 0} \frac{\tan x}{x} = 1$

TRIGONOMETRY LAWS AND IDENTITIES

TANGENT IDENTITIES

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

$$\cot\theta = \frac{\cos\theta}{\sin\theta}$$

RECIPROCAL IDENTITIES

$$\csc\theta = \frac{1}{\sin\theta} \qquad \sin\theta = \frac{1}{\csc\theta}$$

$$\sec\theta = \frac{1}{\cos\theta} \qquad \cos\theta = \frac{1}{\sec\theta}$$

$$\cot\theta = \frac{1}{\tan\theta} \qquad \tan\theta = \frac{1}{\cot\theta}$$

PYTHAGOREAN IDENTITIES

$$\sin^2\theta + \cos^2\theta = 1$$

$$\tan^2\theta + 1 = \sec^2\theta$$

$$\cot^2\theta + 1 = \csc^2\theta$$

PERIODIC IDENTITIES

$$\sin(\theta + 2\pi n) = \sin\theta$$

$$\cos(\theta + 2\pi n) = \cos\theta$$

$$\tan(\theta + \pi n) = \tan\theta$$

$$\csc(\theta + 2\pi n) = \csc\theta$$

$$\sec(\theta + 2\pi n) = \sec\theta$$

$$\cot(\theta + \pi n) = \cot\theta$$

EVEN/ODD IDENTITIES

$$\sin(-\theta) = -\sin\theta$$

$$\cos(-\theta) = \cos\theta$$

$$\tan(-\theta) = -\tan\theta$$

$$\csc(-\theta) = -\csc\theta$$

$$\sec(-\theta) = \sec\theta$$

$$\cot(-\theta) = -\cot\theta$$

DOUBLE ANGLE IDENTITIES

$$\sin(2\theta) = 2\sin\theta\cos\theta$$

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta$$

$$= 2\cos^2\theta - 1$$

$$= 1 - 2\sin^2\theta$$

$$\tan(2\theta) = \frac{2\tan\theta}{1 - \tan^2\theta}$$

HALF ANGLE IDENTITIES

$$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{2}}$$

$$\cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1+\cos\theta}{2}}$$

$$\tan\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{1+\cos\theta}}$$

LAW OF COSINES

$$a^2 = b^2 + c^2 - 2bc\cos\alpha$$

$$b^2 = a^2 + c^2 - 2ac\cos\beta$$

$$c^2 = a^2 + b^2 - 2ab\cos\gamma$$

PRODUCT TO SUM IDENTITIES

$$\sin\alpha\sin\beta = \frac{1}{2}[\cos(\alpha-\beta) - \cos(\alpha+\beta)]$$

$$\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha-\beta) + \cos(\alpha+\beta)]$$

$$\sin\alpha\cos\beta = \frac{1}{2}[\sin(\alpha+\beta) + \sin(\alpha-\beta)]$$

$$\cos\alpha\sin\beta = \frac{1}{2}[\sin(\alpha+\beta) - \sin(\alpha-\beta)]$$

SUM TO PRODUCT IDENTITIES

$$\sin\alpha + \sin\beta = 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$

$$\sin\alpha - \sin\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$$

$$\cos\alpha + \cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$

$$\cos\alpha - \cos\beta = -2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$$

LAW OF SINES

$$\frac{\sin\alpha}{a} = \frac{\sin\beta}{b} = \frac{\sin\gamma}{c}$$

LAW OF TANGENTS

$$\frac{a-b}{a+b} = \frac{\tan\left[\frac{1}{2}(\alpha-\beta)\right]}{\tan\left[\frac{1}{2}(\alpha+\beta)\right]}$$

$$\frac{b-c}{b+c} = \frac{\tan\left[\frac{1}{2}(\beta-\gamma)\right]}{\tan\left[\frac{1}{2}(\beta+\gamma)\right]}$$

$$\frac{a-c}{a+c} = \frac{\tan\left[\frac{1}{2}(\alpha-\gamma)\right]}{\tan\left[\frac{1}{2}(\alpha+\gamma)\right]}$$

SUM/DIFFERENCES IDENTITIES

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$$

MOLLWEIDE'S FORMULA

$$\frac{a+b}{c} = \frac{\cos\left[\frac{1}{2}(\alpha-\beta)\right]}{\sin\left(\frac{1}{2}\gamma\right)}$$

COFUNCTION IDENTITIES

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta$$

$$\csc\left(\frac{\pi}{2} - \theta\right) = \sec\theta$$

$$\tan\left(\frac{\pi}{2} - \theta\right) = \cot\theta$$

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta$$

$$\sec\left(\frac{\pi}{2} - \theta\right) = \csc\theta$$

$$\cot\left(\frac{\pi}{2} - \theta\right) = \tan\theta$$

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θ, x, α, β → सबसे

1 − cos x

cos 2x = 1 − 2 sin² x

cos x/2 = 1 − 2 sin² x/2

cos x = 1 − 2 sin² 1/2 x

2 sin² 1/2 x = 1 − cos x

1 − cos x / sin x

1 − cos 2x / x²

cos 2x = 1 − 2 sin² x

2 sin² x = 1 − cos 2x

Sinat Tiitimeva

α	I					II				III				IV			
	0°	30°	45°	60°	90°	120°	135°	150°	180°	210°	225°	240°	270°	300°	315°	330°	360°
	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{5\pi}{4}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{7\pi}{4}$	$\frac{11\pi}{6}$	2π
$\sin\alpha$	0	$\frac{1}{2}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}\sqrt{3}$	1	$\frac{1}{2}\sqrt{3}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{2}\sqrt{2}$	$-\frac{1}{2}\sqrt{3}$	-1	$-\frac{1}{2}\sqrt{3}$	$-\frac{1}{2}\sqrt{2}$	$-\frac{1}{2}$	0
$\cos\alpha$	1	$\frac{1}{2}\sqrt{3}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{2}\sqrt{2}$	$-\frac{1}{2}\sqrt{3}$	-1	$-\frac{1}{2}\sqrt{3}$	$-\frac{1}{2}\sqrt{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}\sqrt{3}$	1
$\tan\alpha$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	td	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	td	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0
$\csc\alpha$	td	2	$\sqrt{2}$	$\frac{2}{3}\sqrt{3}$	1	$\frac{2}{3}\sqrt{3}$	$\sqrt{2}$	2	td	-2	$-\sqrt{2}$	$-\frac{2}{3}\sqrt{3}$	-1	$-\frac{2}{3}\sqrt{3}$	$-\sqrt{2}$	-2	td
$\sec\alpha$	1	$\frac{2}{3}\sqrt{3}$	$\sqrt{2}$	2	td	-2	$-\sqrt{2}$	$-\frac{2}{3}\sqrt{3}$	-1	$-\frac{2}{3}\sqrt{3}$	$-\sqrt{2}$	-2	td	2	$\sqrt{2}$	$\frac{2}{3}\sqrt{3}$	-1
$\cot\alpha$	td	$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	0	$-\frac{1}{\sqrt{3}}$	-1	$-\sqrt{3}$	td	$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	1	$-\frac{1}{\sqrt{3}}$	-1	$-\sqrt{3}$	td

LATIHAN SOAL – LIMIT TRIGONMETRI

Carilah nilai dari limit berikut:

1. $\lim_{x \to 0} \frac{2 \sin 4x}{7 \tan 7x}$

$= \frac{2}{7} \cdot \lim_{x \to 0} \frac{\sin 4x}{\tan 7x} = \frac{2}{7} \cdot \frac{4}{7} = \frac{8}{49}$ //

$\frac{\sin 4x}{\tan 7x} = \frac{1}{4} \lim_{x \to 0} \frac{\sin 4x}{\tan 7x}$

2. $\lim_{x \to \frac{\pi}{2}} (\sin x + \cos x)$

$= \sin \frac{\pi}{2} + \cos \frac{\pi}{2}$

$= 1 + 0$

$= 1$ //

LATIHAN SOAL – LIMIT TRIGONMETRI

Carilah nilai dari limit berikut:

3. $\lim_{x\to 0} \frac{2x^2+x}{\sin x} = \frac{2(0)^2+0}{\sin 0} = \frac{0}{0}$

4. $\lim_{x\to 0} \frac{\tan 2x \cdot \tan 3x}{3x^2} = \frac{\tan 2(0)\cdot \tan 3(0)}{3(0)^2} = \frac{0}{0}$

5. $\lim_{x\to 0} \frac{1-\cos x}{\sin x} = \frac{1-\cos(0)}{\sin(0)} = \frac{1-1}{0} = \frac{0}{0}$

3. $\lim_{x\to 0} \frac{2x^2+x}{\sin x} = \lim_{x\to 0} \frac{x(2x+1)}{\sin x}$

$= \lim_{x\to 0} \frac{x}{\sin x} \cdot \lim_{x\to 0} 2x+1$

$= 1 \cdot (2(0)+1)$

$= 1 \cdot 1$

$= 1$ //

4. $\lim_{x\to 0} \frac{\tan 2x \cdot \tan 3x}{3x^2}$

$= \lim_{x\to 0} \frac{\tan 2x \cdot \tan 3x}{3x \cdot x}$

$= \lim_{x\to 0} \frac{\tan 2x}{3x} \cdot \lim_{x\to 0} \frac{\tan 3x}{x}$ (→ m, → n)

$= \frac{2}{3} \cdot \frac{3}{1} = \frac{2}{1} = 2$ //

$= \frac{6}{3}$

$= 2$ //

5. $\lim_{x\to 0} \frac{1-\cos x}{\sin x} = \lim_{x\to 0} \frac{2\sin^2 \frac{1}{2}x}{\sin x}$

$\frac{2}{0}$ = tak ada

$= \lim_{x\to 0} \frac{2 \cdot \sin \frac{1}{2}x \cdot \sin \frac{1}{2}x}{\sin x}$ (pembilang 3)

$= \lim_{x\to 0} 2 \cdot \lim_{x\to 0} \frac{\sin \frac{1}{2}x}{\sin x} \cdot \lim_{x\to 0} \sin \frac{1}{2}x$ (→ m, → n)

$= 2 \cdot \frac{\frac{1}{2}}{1} \cdot \sin \frac{1}{2}(0)$ → $\sin 0 = 0$

$= 0$ //

✓ $\lim_{x\to 0} \frac{2x^2+x}{\sin x} = \lim_{x\to 0} \frac{2x^2}{\sin x} + \lim_{x\to 0} \frac{x}{\sin x}$

$= \lim_{x\to 0} \frac{2x \cdot x}{\sin x} + \lim_{x\to 0} \frac{x}{\sin x}$

$= \lim_{x\to 0} 2x \cdot \lim_{x\to 0} \frac{x}{\sin x} + \lim_{x\to 0} \frac{x}{\sin x}$

$= 2(0) \cdot 1 + 1$

$= 0 + 1$

$= 1$ //

$\frac{1+3}{4} = \frac{1}{4} + \frac{3}{4}$

$\frac{1 \cdot 3}{4} = \frac{1}{4} \cdot 3 = \frac{3}{4}$

$= 1 \cdot \frac{3}{4} = \frac{3}{4}$

LATIHAN SOAL – LIMIT TRIGONMETRI

Carilah nilai dari limit berikut:

6. $\lim_{x\to 0} \frac{\cos 2x}{\sin x - \cos x}$ $= \frac{\cos 2(0)}{\sin 0 - \cos 0} = \frac{1}{0-1} = \frac{1}{-1} = -1$ //

7. $\lim_{x\to \frac{\pi}{6}} \frac{1}{2}\cos 12x$ $= \frac{1}{2} \cdot \cos 2(\frac{\pi}{6}) = \frac{1}{2} \cdot \cos 2\pi = \frac{1}{2} \cdot 1 = \frac{1}{2}$ //

8. $\lim_{x\to 0} \frac{2x \sin 3x}{\tan^2 5x}$ $= \frac{2(0)\cdot \sin 3(0)}{\tan^2 5(0)} = \frac{0}{0}$

9. $\lim_{x\to 0} \frac{1-\cos 2x}{x^2}$ $= \frac{1-\cos 2(0)}{0^2} = \frac{1-1}{0} = \frac{0}{0}$

(8) $\lim_{x\to 0} \frac{2x \cdot \sin 3x}{\tan^2 5x} = \lim_{x\to 0} \frac{2x \cdot \sin 3x}{\tan 5x \cdot \tan 5x}$

$= \lim_{x\to 0} \frac{2x}{\tan 5x} \cdot \lim_{x\to 0} \frac{\sin 3x}{\tan 5x}$

$= \frac{2}{5} \cdot \frac{3}{5}$

$= \frac{6}{25}$ //

(9) $\lim_{x\to 0} \frac{1-\cos 2x}{x^2} = \lim_{x\to 0} \frac{2\sin^2 x}{x^2}$

$= \lim_{x\to 0} \frac{2 \cdot \sin x \cdot \sin x}{x \cdot x}$

$= \lim_{x\to 0} 2 \cdot \lim_{x\to 0} \frac{\sin x}{x} \cdot \lim_{x\to 0} \frac{\sin x}{x}$

$= 2 \cdot 1 \cdot 1$

$= 2$ //



Turunan

Pertemuan Ke-6

TURUNAN

derivatif / diferensial

Definisi

Turunan fungsi f adalah fungsi lain f' (dibaca f aksen) yang nilainya pada sebarang bilangan c adalah

$f(x) = x$ $f(x) = -30x$
$f'(x) = 1$ $f'(x) = -30$

turunan → definisi limit

$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}$$

$f(x) = 2x$
$f'(x) = 2$

Aturan Turunan

1. Aturan konstanta. Jika $f(x) = k$, maka $f'(x) = 0$ $f(x) = 5$ $f'(x) = 0$ $f(x) = -100$ $f'(x) = 0$
2. Aturan penjumlahan. Jika ada $f(x)$ dan $g(x)$, maka turunannya $f'(x) + g'(x)$
3. Aturan pangkat. Jika $f(x) = x^n$, maka $f'(x) = nx^{n-1}$ $f(x) = x^6$ $f'(x) = 6x^{6-1} = 6x^5$ // $f(x) = 2x^3$ $f'(x) = 2 \cdot 3x^{3-1} = 6x^2$
4. Aturan perkalian. Jika ada perkalian dua fungsi maka $(uv)' = u'v + uv'$

f(x) g(x) misal

5. Aturan pembagian. Jika ada pembagian dua fungsi maka $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$

LATIHAN SOAL

$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}$$

$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

Carilah turunan berdasarkan definisi dari fungsi berikut:

1. $f(x) = 13x - 6$
2. $f(x) = x^2 - 2x$
3. $f(x) = \frac{1}{x}$
4. $f(x) = \sqrt{x}$

① $f(x) = 13x - 6$

$f(x+h) = 13(x+h) - 6 = 13x + 13h - 6$

$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h \to 0} \frac{(13x + 13h - 6) - (13x - 6)}{h}$

$= \lim_{h \to 0} \frac{13x + 13h - 6 - 13x + 6}{h}$

$= \lim_{h \to 0} \frac{13h}{h}$

$= \lim_{h \to 0} 13 \rightarrow 13$

$= 13$ //

② $f(x) = x^2 - 2x$

$f(x+h) = (x+h)^2 - 2(x+h)$

$= x^2 + 2xh + h^2 - 2x - 2h$

$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h \to 0} \frac{(x^2 + 2xh + h^2 - 2x - 2h) - (x^2 - 2x)}{h}$

$= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - 2x - 2h - x^2 + 2x}{h}$

$= \lim_{h \to 0} \frac{2xh + h^2 - 2h}{h}$

$= \lim_{h \to 0} \frac{h(2x + h - 2)}{h}$

$= \lim_{h \to 0} 2x + h - 2$

$= 2x + 0 - 2 \quad = 2x - 2$ //

③ $f(x) = \frac{1}{x}$

$f(x+h) = \frac{1}{x+h}$

$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h\to 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{\frac{h}{1}}$

$= \lim_{h\to 0} \left(\frac{1}{x+h} - \frac{1}{x}\right) \cdot \frac{1}{h}$

$= \lim_{h\to 0} \left(\frac{x - (x+h)}{(x+h)(x)}\right) \cdot \frac{1}{h}$

$= \lim_{h\to 0} \left(\frac{x - x - h}{(x+h)(x)}\right) \cdot \frac{1}{h}$

$= \lim_{h\to 0} \frac{-h}{(x+h)(x)(h)}$ $-1h$

$= \lim_{h\to 0} \frac{-1}{(x+h)(x)}$

$= -\frac{1}{(x+0)(x)} = -\frac{1}{x^2}$ //

④ $f(x) = \sqrt{x}$

$f(x+h) = \sqrt{x+h}$

$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h\to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}}$ $-\sqrt{x}\cdot\sqrt{x}$

$= \lim_{h\to 0} \frac{(x+h) + \sqrt{x}\sqrt{x+h} - \sqrt{x}\sqrt{x+h} - x}{h(\sqrt{x+h} + \sqrt{x})}$

$= \lim_{h\to 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})}$

$= \lim_{h\to 0} \frac{1}{\sqrt{x+h} + \sqrt{x}}$

$= \frac{1}{\sqrt{x+0} + \sqrt{x}}$

$= \frac{1}{\sqrt{x} + \sqrt{x}}$

$= \frac{1}{2\sqrt{x}}$ //

LATIHAN SOAL

Carilah turunan dari fungsi berikut:

1. $f(x) = x^3 + 3x^2 + 4$

2. $f(x) = (3x^2 - 5)(2x^4 - x)$

3. $f(x) = \frac{x+3}{x^2+1}$

4. $f(x) = \frac{5}{\sqrt{x}}$

5. $f(x) = \frac{3}{x^3} + x^{-4}$

Thank you

